PARISHRAM 2.0

CLASS 12TH 2026

d & f-Block

CHEMISTRY | LECTURE-5

BY SANYA NAYAR MA'AM



TOPICS TO BE COVERED

1. Potassium Permanganate




ROAR
Tiger hai tu Tiger! Roar

Chemical Reactivity and E° Values

Transition metals vary widely in their chemical reactivity.

- Metals of 1st series (3d) are relatively more reactive (exception Cu) and are oxidized by 1 M H^+.

$$\overset{0}{Zn} + \text{dil HCl} \longrightarrow \overset{+2}{Zn}Cl_2 + H_2\uparrow$$

oxidation

Cu + HCl → X

Note

Ti and V: passive to dilute non-oxidizing acids at room temperature.

↳ HCl

- E° Values for M^{2+}/M : indicate a ↓sing tendency to form divalent (+2) cations across the series. This is related to the ↑se in sum of 1st and 2nd I.E.

K Na Ca Mg Al Zn Fe Pb H Cu Hg Ag Au Pt

Chemical Reactivity and E° Values

- E° values for Mn^{2+}/Mn, Ni^{2+}/Ni and Zn^{2+}/Zn are more –ve than expected.

  Mn^{2+} (d^5) and Zn^{2+} (d^{10}) –> due to half filled and fully filled E.C

  Ni^{2+}: E° value is related to highest –ve $\Delta H_{hydration}$.

Mn^{2+} Co^{2+}

» Mn^{3+} and Co^{3+}: strong oxidizing agents in aq. solution.

» Ti^{2+}, V^{2+} and Cr^{2+} : strong reducing agents and liberate H_2 from a dil. acid.

e.g. $2Cr^{2+}(aq) + 2H^+(aq) \rightarrow 2Cr^{3+} + H_2(g)$

$3d^4$ $3d^3$



Half filled stable t_{2g} configuration

QUESTION



How would you account for the irregular variation of ionisation enthalpies (first and second) in the first series of the transition elements?

In 3d Series, the irregularity of first and second I.E is due to

-irregularity in size

-irregularity in electronic configuration

QUESTION



Why is the highest oxidation state of a metal exhibited in its oxide or fluoride only?

This is because fluorine and oxygen are highly electronegative elements and can oxidise metals to higher oxidation states

QUESTION

Which is a stronger reducing agent Cr^{2+} or Fe^{2+} and why?



$E^{\circ}\ Cr^{3+}/Cr^{2+} = -0.41V$

$E^{\circ}\ Fe^{3+}/Fe^{2+} = +0.77V$

Cr^{2+} is a stronger R.A

More -ve E° = Better R.A

More +ve E° = Better O.A

QUESTION



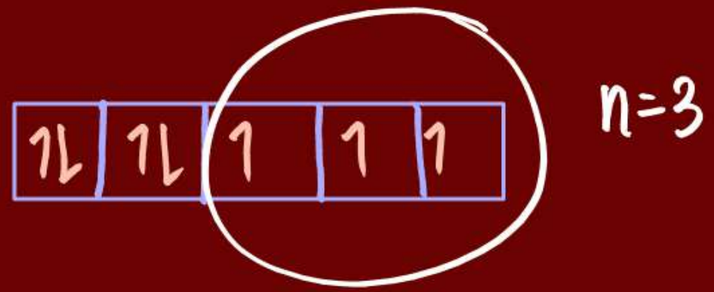
Calculate the 'spin only' magnetic moment of M^{2+} (aq) ion (Z = 27).

$\mu_s = \sqrt{n(n+2)}$ n = no. of unpaired e⁻s.

$\mu_s = \sqrt{3(3+2)} = \sqrt{15} = 3.87\ BM$

$_{27}M : 3d^7 4s^2$

$M^{2+} : 3d^7$



Potassium permanganate ($KMnO_4$)

Preparation :

MnO_2

i) It is prepared from pyrolusite ore with KOH in the presence of oxidising agent like KNO_3. The dark green potassium manganate undergoes electrolytic oxidation to produce potassium permanganate.

$KNO_3 \xrightarrow{\Delta} KNO_2 + O_2$

$$\overset{+4}{2MnO_2} + 4KOH + O_2 \rightarrow 2K_2\overset{+6}{Mn}O_4 + 2H_2O$$

Pyrolusite ore — Potassium Manganate

$$3\overset{+6}{Mn}O_4^{2-} + 4H^+ \rightarrow 2\overset{+7}{Mn}O_4^- + MnO_2 + 2H_2O$$

Manganate ion — Permanganate ion

Potassium permanganate ($KMnO_4$)

Preparation :

ii) Commercially, it is prepared by alkaline oxidative fusion of MnO_2 followed by electrolytic oxidation of manganate (VI).

$$\overset{+4}{MnO_2} + 2e^- \xrightarrow{\text{Fused with KOH \& oxidised with air/KNO}_3} \underset{\text{Manganate ion}}{\overset{+6}{MnO_4^{2-}}}$$

$$\underset{\text{Manganate ion}}{MnO_4^{2-}} \xrightarrow{\text{Electrolytic oxidation in alkaline solution}} \underset{\text{Permanganate ion}}{MnO_4^-} + 1e^-$$

Potassium permanganate ($KMnO_4$)

Preparation :

iii) In laboratory, by oxidation of manganese (II) ion salt by peroxodisulphate

$$2Mn^{2+} + 5S_2O_8^{2-} + 8H_2O \rightarrow 2\overset{+7}{M}nO_4^- + 10SO_4^{2-} + 16H^+$$

Peroxodisulphate

$H_2S_2O_8$: Peroxodisulphuric acid

Potassium permanganate ($KMnO_4$)

Properties:

i) Dark purple needle like crystalline solid with metallic lustre.

ii) Sparingly soluble in water

iii) Decomposes on heating at 513 K.

$$2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$$

iv) Acts as a powerful oxidizing agent in acidic, alkaline and neutral medium

Acidic: $Mn^{+7} \rightarrow Mn^{+2}$

Basic: $Mn^{+7} \rightarrow Mn^{+6}$

Neutral: $Mn^{+7} \rightarrow Mn^{+4}$

Potassium permanganate ($KMnO_4$)

In acidic medium $KMnO_4$ oxidises:

i) **Iodide to iodine**

Reduction Half Rxn: $\overset{+7}{MnO_4^-} + 5e^- + 8H^+ \longrightarrow Mn^{+2} + 4H_2O$] X2

Oxidation Half Rxn: $2I^- \longrightarrow I_2 + 2e^-$] X5

Overall Rxn: $2MnO_4^- + 10I^- + 16H^+ \longrightarrow 2Mn^{+2} + 8H_2O + 5I_2$

Potassium permanganate ($KMnO_4$)

In acidic medium $KMnO_4$ oxidises:

(ii) Ferrous to Ferric

Reduction Half Rxn

$$MnO_4^- + 5e^- + 8H^+ \longrightarrow Mn^{2+} + 4H_2O$$

Oxidation Half Rxn

$$Fe^{2+} \longrightarrow Fe^{3+} + 1e^- \quad] \times 5$$

Overall Rxn

$$MnO_4^- + 8H^+ + 5Fe^{2+} \longrightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$$

Potassium permanganate ($KMnO_4$)

In acidic medium $KMnO_4$ oxidises:

iii)Oxalate to carbon dioxide

Reduction Half Rxn

$$MnO_4^- + 5e^- + 8H^+ \longrightarrow Mn^{2+} + 4H_2O \quad]\times 2$$

Oxidation Half Rxn

$$C_2O_4^{2-} \longrightarrow 2CO_2 + 2e^- \quad]\times 5$$

Overall Rxn

$$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \longrightarrow 2Mn^{2+} + 8H_2O + 10CO_2$$

Potassium permanganate ($KMnO_4$)

In acidic medium $KMnO_4$ oxidises:

vi) **Hydrogen sulphide to sulphur**

Reduction Half Rxn: $MnO_4^- + 5e^- + 8H^+ \longrightarrow Mn^{2+} + 4H_2O$] x2

Oxidation Half Rxn: $S^{2-} \longrightarrow S + 2e^-$] x5

Overall Rxn: $2MnO_4^- + 16H^+ + 5S^{2-} \longrightarrow 2Mn^{2+} + 8H_2O + 5S$

Potassium permanganate ($KMnO_4$)

1. **In acidic medium $KMnO_4$ oxidises:**

   v) **Sulphite to sulphate**

Reduction Half Rxn

$$MnO_4^- + 5e^- + 8H^+ \longrightarrow Mn^{2+} + 4H_2O \quad] \times 2$$

Oxidation Half Rxn

$$SO_3^{2-} + H_2O \longrightarrow SO_4^{2-} + 2e^- + 2H^+ \quad] \times 5$$

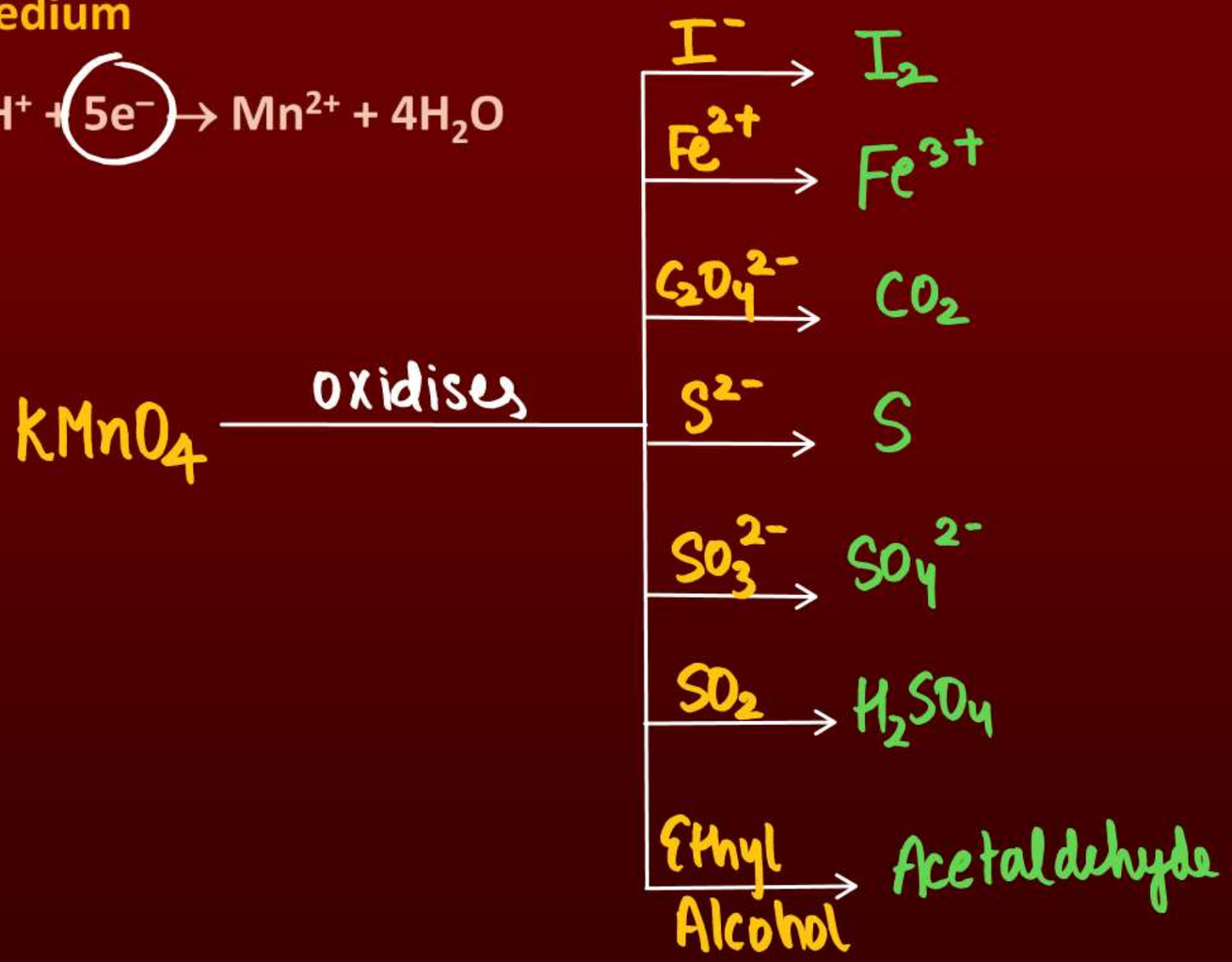
Overall Rxn

$$2MnO_4^- + 5SO_3^{2-} + 6H^+ \longrightarrow 2Mn^{2+} + 5SO_4^{2-} + 3H_2O$$

Potassium permanganate ($KMnO_4$)

1. **In acidic medium**

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$



Eq.wt. of $KMnO_4$ in acidic medium

$$\text{Eq.wt.} = \frac{\text{Mol.wt.}}{\text{n-factor}}$$

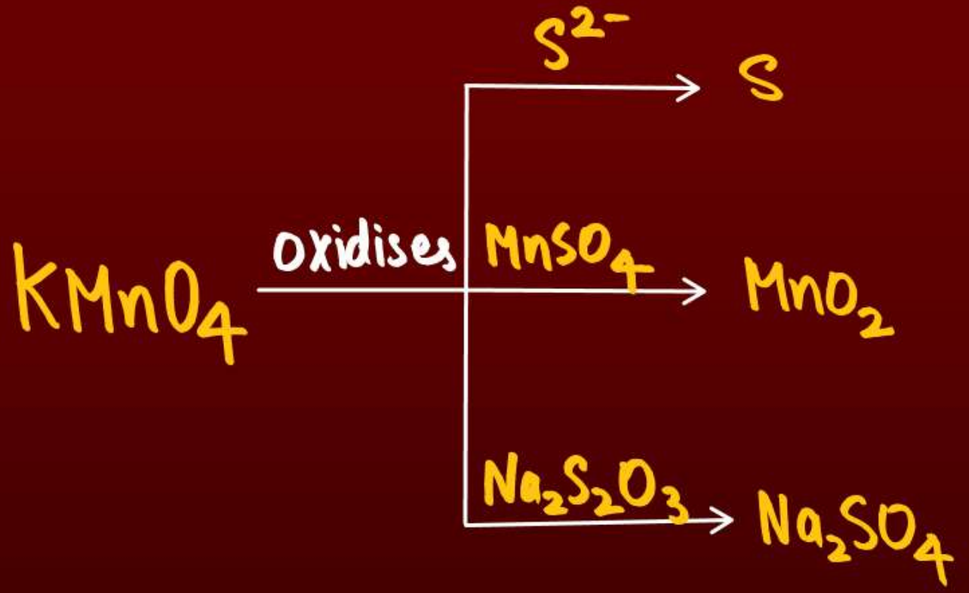
$$= \frac{158}{5} = 31.6$$

Potassium permanganate ($KMnO_4$)

2. In neutral medium

$$MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$$

$$\text{Eq. wt.} = \frac{158}{3} = 52.67$$

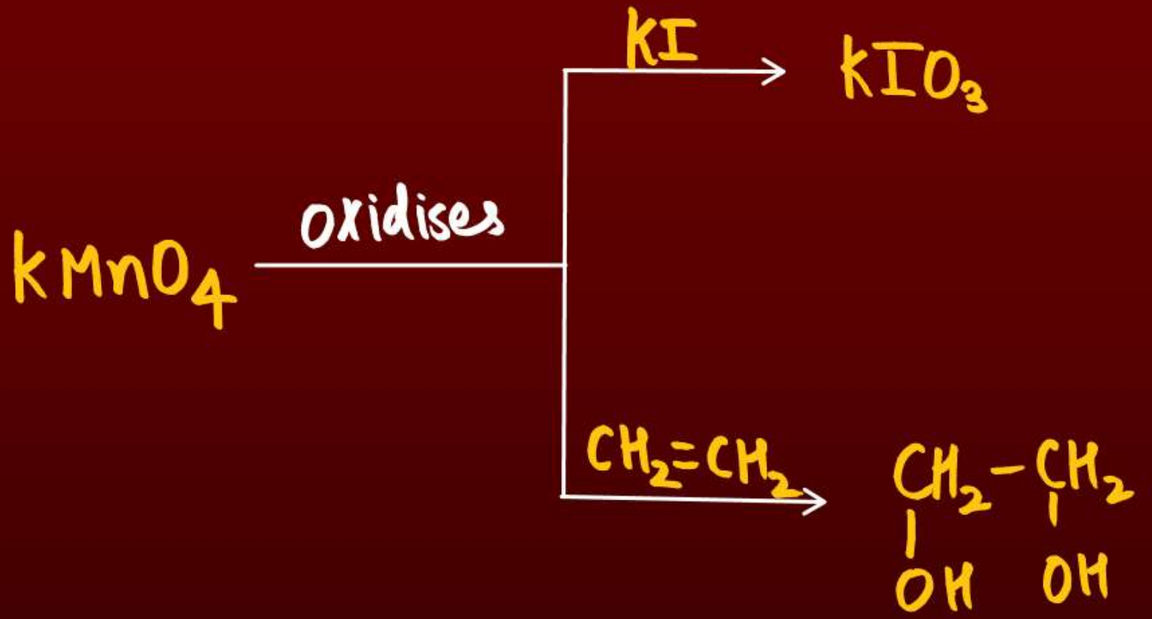


Potassium permanganate ($KMnO_4$)

3. In alkaline medium

$$2KMnO_4 + 2KOH \rightarrow 2K_2MnO_4 + H_2O + [O]$$

$$\text{Eq. wt.} = \frac{158}{1} = 158$$



Uses of $KMnO_4$

i) Bleaching of wool, silk, cotton and other textile fibres etc.

ii) Decolourisation of oils.

iii) In analytical chemistry (titration).

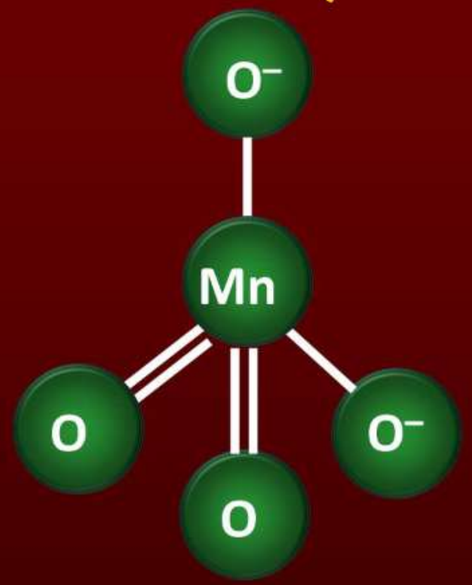
iv) In organic synthesis.

v) As disinfectant of water and used in dry cells as well.

Uses of $KMnO_4$

Structure :

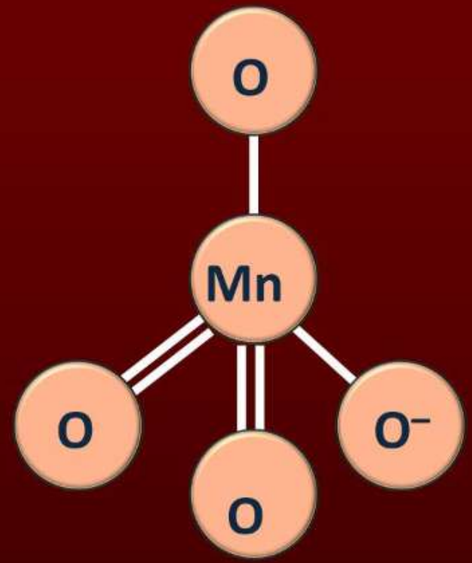
$^{+6}MnO_4^{2-}$ (Manganate ion)



Tetrahedral Manganate Ion

(Green)

$^{+7}MnO_4^-$ (Permanganate ion)



Tetrahedral Permanganate Ion

(Purple)

Homework

→ DPP Solve

→ NCERT Read + Q's.

THANK YOU